

ZUHAB WASIM

SKILLS

 Python

 C#

 Java

 JavaScript

 PostgreSQL

 GLSL

TOOLS

 AWS

 Docker

 Unity

 Godot

LIBRARIES

 Pydantic

 FastAPI

 asyncio

 NumPy

 React

 Three.js

 Open3D

INFO

 zuhabwasim.github.io

 /ZuhabWasim

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EXPERIENCE

3D Software Engineer

iCAD Dental

Oct 2024—Present

- Refactored a 10k-line monolith into 10 Dockerized FastAPI microservices, slashing technical debt and enabling stateless AWS deployment
- Standardized PEP8 guides, microservice structure, PR templates, Git-JIRA workflows, Swagger docs, presenting updates directly to C-level stakeholders
- Vectorized ray generation and removed costly tensor/NumPy conversions to boost runtime by 1000x (0.5s to 0.0005s)
- Compressed payload 80 MB responses to 2 MB using MsgPacks, zstd and clever JSON structuring
- Rebuilt mesh-alignment algorithm with a convex-hull + DFS approach, cutting 5s runtime <1s ensuring deterministic and reliable transforms

Software Developer

BiblioCommons

May 2020—Aug 2021

- Integrated modern web, SaaS infrastructure for libraries globally, managing client onboarding, testing, PagerDuty and troubleshooting
- Decoupled local client systems favouring AWS EC2 cloud solutions, optimizing nightly database syncs to halving loading time
- Migrated 1+ million records for a large client with Python, Java, and PostgreSQL with no downtime or data loss, earning “star” employee accolades

PROJECTS

3D Tooth Analytics Backend – iGDS

Python FastAPI Open3D NumPy Blender

Oct 2024—Mar 2025

- Engineered over/under-prep analytics with KD-Trees / Open3D raycasting to aid dentists chairside
- Designed abutment-angle algorithm using Savitzky–Golay smoothing and PCA/SVD regression applying GPU-accelerated computing
- Derived new 3D "sidewall-base" feature reaching sub-millimetre accuracy for undercut algorithms

Visualizer Engine - Dental UI

Three.js WebGL GLSL React MsgPack Facade Pattern

Jan 2025—May 2025

- Modularized a Three.js visualizer into manager classes with a facade API for easy UI links, dividing 1000+ line monolith into <200 line production-ready modules
- Built custom GLSL shaders with Fresnel x-rays, data textures for dental preps, using barycentrics to get 2x faster performance than built-in options

Reflection - Game

C# Unity Git

Jan 2022—Apr 2022

- Led a 10-person cross-disciplinary to design a 30-minute narrative/puzzle campaign in 3 months
- Contributed >50% of C# backend game logic, becoming the course's most polished game, reaching 600+ views and 150+ downloads at launch

EDUCATION

University of Toronto

Honours Bachelor in Computer Science, Minor in Mathematics

2022

AI / ML Databases Operating Systems SOLID Principles Algorithms